

De Finetti's Gambit and the Construction of Chance

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Abstract. A theorem due to Bruno de Finetti convinced him and many of his followers that we don't need objective chances in our physics or metaphysics. In its standard exposition, this view is a radical form of error theory. I develop a more nuanced view according to which chance plays an important role in our representation of the world, without directly tracking any objective quantity. I argue that this kind of approach might provide an attractive alternative to more familiar developments of antirealism across philosophical domains.

1 Introduction

Objective chance is puzzling. Hypotheses about chance seem to be empirically inert: they entail nothing about actual outcomes; yet information about outcomes closely guides our beliefs about chance. Faced with such puzzles, some have advanced “personalist” or “projectivist” accounts, according to which chance is not an objective aspect of the world, but a mere projection from our habits or degrees of belief.¹ Bruno de Finetti [1931; 1970] gave mathematical substance to this idea. His celebrated representation theorem appears to show that a personalist account of chance can make sense of our scientific practice, in a way that no realist account of chance can. As Persi Diaconis and Brian Skyrms [2017: p.123] put it: “de Finetti showed [...] that if we dispense with objective chance, *nothing is lost*. The mathematics of inductive reasoning remains exactly the same.”

De Finetti's theory of chance has been endorsed by an impressive list of philosophers, scientists, and statisticians.² Unfortunately, presentations of the theory tend to be heavy on the maths and light on the philosophy. De Finetti's theorem is a mathematical fact, a consequence of the probability calculus. How do we get from this to a philosophical theory of chance? And what, exactly, is that theory? How are utterances and thoughts about chance to be understood, if there are no chances out

¹ See, among many others, [Hume 1739: bk.I pt.III sec.XI], [Ramsey 1931: 206–211], [Jeffrey 1983: sec.12.7], [Blackburn 1983], [Logue 1995], [Ward 2002].

² See, among others, [Skyrms 1984: ch.3], [Jeffrey 2004], [Bernardo and Smith 2009], [Poirier 2011], [Caves et al. 2002], [Fuchs and Schack 2015], [Diaconis and Skyrms 2017: ch.7], [Spiegelhalter 2024]. See also [Gillies 2000: ch.4] and [Galavotti 2005: ch.7] for comparisons with other accounts.

there in the world?

I will outline an answer to these questions that goes beyond the simplistic antirealism often associated with de Finetti's program. On the account I will develop, chance is an extra parameter in our model of the world: it serves a useful role in our cognitive architecture, without representing an objective element of reality.

In developing this account of chance, I hope to put a new form of antirealism on the philosophical map. For the last 100 years or so, antirealists about colour, ethics, mathematics etc. have generally focused on language: on how we should interpret statements and utterances that appear committed to the problematic features of reality. I suggest that we begin with (Bayesian) epistemology and psychology. If our cognitive model of the world involves extra parameters for colours, norms, chances, etc., it is no surprise that we may use declarative sentences to express beliefs involving these parameters. The facts about language naturally fall into place.

2 Credence and Chance

Before we begin, let me say a few words on why one might seek an antirealist account of chance, and what I mean by 'chance'. When I say 'chance', I mean the kind of probability that figures in scientific theories: in the collapse law of GRW quantum mechanics, the laws of radioactive decay, the Maxwell-Boltzmann law of molecular velocity, or the Gutenberg-Richter law of earthquakes. Probabilistic theories are ubiquitous in science, but there is no agreement on what they mean. What does it mean to say that a coin has a 50% chance of landing heads, or that a radium-227 atom has a 50% probability of decaying within 42 minutes?

It is clear that claims like these can't be directly understood as statements about belief. If we're unsure about the half-life of radium-227, we're not thereby unsure about our own beliefs, or about anyone else's beliefs. A naive personalist interpretation of chance would turn most of science into a branch of psychology. This will not do.

Science textbooks often define probability as relative frequency: "the number of favourable cases divided by the number of all the cases", as Laplace [1814] put it. This can't be right either. It would make it logically impossible that a fair coin is tossed an odd number of times. Switching to limiting relative frequencies in hypothetical trials, as some textbooks suggest, arguably makes things worse. It is doubtful that there is a fact of the matter about how a coin would land if it were tossed an infinite number of times. Besides, any world where this happens would be quite unlike ours. Why should the science of our world be obsessed with what happens in remote worlds like that?³

If chance can't be defined in terms of anything else, perhaps it should be taken as primitive. According to non-reductive realism about chance, the physical universe contains fundamental probabilistic quantities; chance statements are simply statements about these quantities.

Here, as elsewhere, non-reductive realism faces objections from metaphysics, epistemology, and the philosophy of language. Metaphysically, chance facts appear to be essentially modal: they don't concern what *is* the case, but what *might* be the case. They also appear to be intentional: the chance of an outcome is somehow *about* that outcome. One might deem a metaphysics with primitively

³ For further arguments against frequentist interpretations, see, e.g., [Hájek 1997] and [Hájek 2009].

modal and intentional elements objectionably queer. One may also wonder how we could have epistemic access to supposedly primitive chance facts, and how our words could latch onto them. Worse, remember that we find probabilistic theories not just in fundamental physics, but also in statistical mechanics, systems biology, population ecology, epidemiology, and many other fields. These theories do not require probabilities at the level of fundamental physics. The chances of coin flips and molecular diffusion aren't physically fundamental. If so, it is hard to see how they could be metaphysically fundamental.

Let's take a closer look at the epistemic problem for realism, as it will lead us to de Finetti's alternative. Consider what a God would have to do in order to create a world with fundamental chances. One thing she would have to do is settle the chances. Imagine she turns a dial to determine the chance of heads for a truly indeterministic type of coin flip. Unless she set the dial to 0 or 1 (and perhaps even then), this leaves open how each individual flip will turn out. To settle this as well, God would have to decide that the first flip lands heads, the second tails, and so on. The two choices are independent: any combination of (non-trivial) chance settings is metaphysically compatible with any sequence of outcomes. (A coin that's biased towards heads can still land tails on every trial.) If we live in a world that has been created in this fashion, how could we get information about the dial setting? How could it help to look at the outcomes, or to scrutinize the coins?

The severity of this problem is easy to miss because we know very well how to reason from outcomes to chances. In its Bayesian regimentation, the inference goes as follows.⁴

Let Cr be the credence function of a rational scientist (whatever exactly this involves). Imagine that our scientist is interested in a certain type of coin flip. She believes that there's a fixed chance θ of heads on each trial, but she is unsure about the value of θ . Then she observes a sequence of outcomes $x_1, \dots, x_n$, with $\#h$ heads and $\#t$ tails. How should this affect her credence towards θ ? *Bayes' Rule* gives the answer:

$$Cr(\theta \mid x_1, \dots, x_n) \propto \theta^{\#h} (1 - \theta)^{\#t} \times Cr(\theta). \quad (BR)$$

The more outcomes the scientist observes, the more her posterior credence is thereby concentrated around $\#h/n$.

(BR) can be derived from Bayes' Theorem (a theorem of the probability calculus), an independence assumption, and a version of the Principal Principle. The independence assumption says that conditional on a particular chance value θ , no outcome provides evidence about any other (compare [Skyrms 1980] on chance and "resiliency"):

$$Cr(x_1, \dots, x_n \mid \theta) = Cr(x_1 \mid \theta) \times \dots \times Cr(x_n \mid \theta). \quad (Ind)$$

The required version of the Principal Principle says that the credence in any heads outcome, conditional on heads having chance θ , equals θ :⁵

$$Cr(x_i = H \mid \theta) = \theta. \quad (PP)$$

⁴ See, e.g., [Howson and Urbach 1993] and [Earman 1992] for general surveys of the Bayesian account of scientific reasoning. While not entirely uncontroversial, it is arguably a presupposition of de Finetti's gambit.

⁵ This is entailed by the Principal Principle of [Lewis 1980] on the assumption that Cr doesn't have any "inadmissible evidence".

I'll assume that an account along these lines accurately captures how one can reason from outcomes to chances. So *how* is not really the problem. The problem is to explain why this method works. Why should we be highly confident that God has set the dial to near $2/3$ just because we observe twice as many heads as tails? Why should we assume that there is any connection at all between the outcomes and the dial setting?

Reductive accounts don't seem to fare much better. If chance is (actual) relative frequency, for example, Bayes' Rule is demonstrably unsound: conditional on a particular hypothesis about relative frequency, each outcome in a finite sequence provides evidence about the others. The same is true for the "Best Systems" interpretation of [Lewis 1994].

This is where de Finetti enters the stage. De Finetti saw that if we understand chance as a projection from our attitudes, we can explain our scientific practice: we can derive Bayes' Rule from a plausible symmetry constraint on rational credence. That's the upshot of de Finetti's representation theorem.

To be clear: I haven't shown that realist accounts of chance are untenable, or that the problems I have raised can't be answered. But my aim in this paper is not to argue for antirealism about chance. My aim is to develop de Finetti's ideas into a sophisticated form of antirealism. Even if we ultimately decline de Finetti's gambit, it is worth probing how it might play out.

3 De Finetti's Theorem

To understand de Finetti's proposal, we need to review de Finetti's Theorem. Let's assume (following de Finetti, and Hume, and Democritus) that the world is nothing but a pattern of local events or "outcomes". To simplify the maths, it will be useful to pretend that these outcomes are linearly ordered and fall into just two types. In short, we'll pretend that the world is a sequence of coin flips.

We're going to investigate credence functions over worlds of this kind – over sequences of heads and tails. As before, we imagine that these are the credences of an idealized scientist. Initially, the scientist doesn't know anything about the "world sequence", not even its length. We assume, however, that she regards the sequence as *exchangeable*, meaning that her credence is invariant under permuting outcomes: HHTHTTHT and THHHTTHT, for example, get the same credence because they contain the same number of heads and tails.

In essence, de Finetti's Theorem states that if Cr is such a credence function, then it can be *represented* as a credence function that

- (i) is unsure about the chance θ of heads on each trial,
- (ii) regards the trials as independent, and
- (iii) satisfies Bayes' Rule and the Principal Principle.

Let's work backwards to see what such a representation would look like. What would our scientist's credence function look like if she satisfied conditions (i)–(iii)? Let $x_1, \dots, x_n$ be a possible sequence of outcomes, where each x_i is either heads or tails. If the sequence $x_1, \dots, x_n$ contains $\#h$ heads and $\#t$ tails, the Principal Principle (PP) and the independence assumption (Ind) entail that the scientist's credence in $x_1, \dots, x_n$, conditional on the assumption that the chance of heads is θ , equals $\theta^{\#h}(1 - \theta)^{\#t}$:

$$Cr(x_1, \dots, x_n \mid \theta) = \theta^{\#h}(1 - \theta)^{\#t}. \quad (1)$$

To get her unconditional credence in $x_1, \dots, x_n$, we use the law of total probability, taking a weighted average over all possible values of θ . Since there are uncountably many such values, this requires integration:

$$\text{Cr}(x_1, \dots, x_n) = \int_0^1 \theta^{\#h} (1 - \theta)^{\#t} \text{Cr}(d\theta). \quad (2)$$

The exact meaning of (2) won't be important. Just keep in mind that this describes the credence function of a “realist” scientist who satisfies conditions (i)–(iii). De Finetti's Theorem shows that even if our scientist does not believe in objective chance, her credence function may be represented in the form of (2). I'll briefly sketch how this can be proved. (Again, the details won't be important.)⁶

To begin, recall that $\text{Cr}(x_1, \dots, x_n)$ is our scientist's credence that the first n outcomes in the “world sequence” are $x_1, \dots, x_n$. The whole world sequence $x_1, \dots, x_N$ might have further elements. We assume that Cr regards the world sequence as exchangeable. Now choose a particular $N > n$, and let R stand for the ratio of heads in $x_1, \dots, x_N$. By the law of total probability,

$$\text{Cr}(x_1, \dots, x_n) = \sum_r \text{Cr}(x_1, \dots, x_n \mid R = r) \text{Cr}(R = r). \quad (3)$$

By exchangeability, all sequences $x_1, \dots, x_N$ with the same number of heads have the same probability. This means that $\text{Cr}(x_1, \dots, x_n \mid R = r)$ is the distribution one gets when drawing n balls from an urn with N white or black balls, with a ratio r of white to black. The larger N is compared to n , the more this distribution resembles a binomial distribution. If the sequence $x_1, \dots, x_n$ contains $\#h$ heads, and the world sequence $x_1, \dots, x_N$ contains $\#H (= rN)$ heads, then there are $\binom{N}{\#H}$ overall possibilities for $x_1, \dots, x_N$, and $\binom{N-n}{\#H-\#h}$ possibilities whose initial segment is $x_1, \dots, x_n$. So

$$\text{Cr}(x_1, \dots, x_n \mid R = r) = \frac{\binom{N-n}{rN-\#h}}{\binom{N}{rN}}. \quad (4)$$

Putting (3) and (4) together, we have

$$\text{Cr}(x_1, \dots, x_n) = \sum_r \frac{\binom{N-n}{rN-\#h}}{\binom{N}{rN}} \text{Cr}(R = r). \quad (5)$$

Assuming that Cr puts no upper bound on the size N of the world, this holds for any $N > n$. As we increase N , the ratio R converges (with probability 1) to a limit θ ; $\binom{N-n}{rN-\#h} / \binom{N}{rN}$ converges to $\theta^{\#h} (1 - \theta)^{\#t}$; the sum becomes an integral, and $\text{Cr}(R = r)$ turns into a (unique) measure μ over $\theta \in [0, 1]$.⁷ That is, there is a unique measure μ such that

$$\text{Cr}(x_1, \dots, x_n) = \int_0^1 \theta^{\#h} (1 - \theta)^{\#t} \mu(d\theta). \quad (6)$$

This is de Finetti's Theorem.

⁶ My proof follows [Diaconis and Freedman 1980] rather than de Finetti, who assumes that the world sequence is infinite.

⁷ This assumes that the probability that Cr assigns to $x_1, \dots, x_n$ does not depend on N : mere information about the size of the world sequence doesn't reveal anything about the first n outcomes. More precisely, I assume infinite extendability and projective consistency, as discussed in [Diaconis and Freedman 1980]. We'll see in section 7 that these subtleties aren't crucial for my eventual proposal.

Now compare (6) and (2). They look exactly the same, except that (2) has $\text{Cr}(d\theta)$ where (6) has $\mu(d\theta)$. In the realist equation (2), $\text{Cr}(d\theta)$ is the scientist's credence over the possible values of the chance parameter θ .⁸ De Finetti's Theorem shows that if a scientist regards the outcomes as exchangeable, it's *as if* she believed that they are generated by a stable chance process. Her credence is representable in the form of (6), wherein the probability measure μ behaves just like a credence function over the relevant chances.

As new information about outcomes arrives, this "implicit credence" evolves by Bayes' Rule. Let's abuse notation and write ' $\text{Cr}(\theta)$ ' and ' $\text{Cr}(\theta \mid x_1, \dots, x_n)$ ' for the measure μ in the de Finetti representation of the credence function Cr before and after conditioning on $x_1, \dots, x_n$, respectively. By Bayes' Theorem,⁹

$$\text{Cr}(\theta \mid x_1, \dots, x_n) \propto \theta^{\#h} (1 - \theta)^{\#t} \times \text{Cr}(\theta).$$

The appeal of de Finetti's gambit should now be clear. On realist accounts of chance, Bayes' Rule was a mystery. We couldn't explain why a rational scientist would update her beliefs about chance in accordance with Bayes' Rule. De Finetti saw that if we sacrifice realism, our scientific practice becomes intelligible. Scientific reasoning, with its apparent commitment to objective chance, doesn't rest on metaphysically and epistemologically dubious foundations. At most, it can be accused of sloppy notation.

I've assumed that the world is a sequence of coin flips. If we drop this assumption, the maths becomes harder. De Finetti's gambit assumes that all scientific reasoning about probability can be derived from relevant symmetries in credences over outcomes. There are some open questions here, but I'll assume that this mathematical part of the project can be completed.¹⁰ In section 7, I'll argue that it doesn't matter much if it can't.

4 The Construction of Chance

At this point, we have a mathematical result and an intriguing idea. We don't yet have a philosophical theory. How are we to understand thought and talk about chance, if there are no chances out there in the world?

One possibility is to reject the concept of chance. On this account, it is OK to reason as if there were objective chances, but there is really no such thing. There is no half-life of radium-227. Molecular velocities do not follow the Maxwell-Boltzmann distribution. There are no diffusion coefficients. There is no chance of winning the lottery. That is: there are no objective truths about these matters. All genuine probabilities are subjective degrees of belief. In this incarnation, de Finetti's gambit is a revolutionary form of error theory, calling for a radical revision of our attitudes. It would seem to be an exaggeration to claim, with Diaconis and Skyrms, that "nothing is lost".

I want to explore a less revolutionary version of de Finetti's gambit. On this account, chance is an

⁸ The ' d ' indicates that this is a measure over a continuous space. Roughly speaking, $\text{Cr}(d\theta)$ is the scientist's credence that the chance of heads lies in a tiny interval around θ .

⁹ I'm ignoring some measure-theoretic subtleties that would require writing ' $\text{Cr}(d\theta)$ ' instead of ' $\text{Cr}(\theta)$ '.

¹⁰ See [Skyrms 1984: ch.3] and [Diaconis and Skyrms 2017: ch.7] for an overview of the required generalizations of de Finetti's Theorem.

important parameter in our model of the world, even if it isn't a parameter of the world itself.¹¹ I will proceed in stages.

As the first step, I suggest re-interpreting the measure μ in de Finetti's representation (6) as the subject's actual credence over chance values. Our scientist isn't just reasoning *as if* she believed in chance. She *does* believe in chance, insofar as her credence function is defined over hypotheses about chance. At this stage, however, the chance parameter θ isn't "free": the credence over θ is determined by the credence over outcomes.¹²

Let me make this more precise. Starting with an exchangeable credence function Cr over an algebra of outcomes, we want to define an extended credence function Cr^+ that is also defined over hypotheses about chance, so that

$$\text{Cr}^+(\text{Ch}(H) \in X) = \mu(X),$$

where μ is the measure in the de Finetti representation of Cr and X is any measurable subset of $[0,1]$.

If that's all we say about Cr^+ , we don't get a well-defined probability function: the domain of Cr^+ won't contain, say, any disjunctions of a chance hypothesis with a hypothesis about outcomes, which is required by standard probability theory. We face a Bayesian version of the Frege-Geach problem.

Fortunately, the problem is easily solved. Let the domain of Cr^+ be the product algebra of the outcome algebra and the chance algebra (the domains of Cr and μ). If these algebras are atomic, the new "worlds" (points) in the domain of Cr^+ consist of the old worlds in the domain of Cr extended by an extra dimension for the chance parameter θ . Now remember that de Finetti's representation gives us not just an implicit credence over chance hypotheses, but also an implicit credence over outcomes conditional on chance:

$$\text{Cr}(x_1, \dots, x_n \mid \theta) = \theta^{\#h} (1 - \theta)^{\#t}.$$

We can therefore define a joint probability over outcomes and chance hypotheses by mirroring the standard rule of probability, that $\text{Cr}(A \wedge B) = \text{Cr}(A \mid B) \text{Cr}(B)$:

$$\text{Cr}^+(x_1, \dots, x_n \wedge \text{Ch}(H) \in X) = \int_X \theta^{\#h} (1 - \theta)^{\#t} \cdot \mu(d\theta).$$

This defines the extended credence function Cr^+ .

The extended function is reducible to the original function Cr insofar as any exchangeable credence Cr over outcomes determines a unique extended credence function Cr^+ . The functions Cr and Cr^+ agree on the original domain of outcomes, but Cr^+ is also defined for pure chance hypotheses, and for arbitrary Boolean combinations of hypotheses about chance and hypotheses about outcomes.

Of course, Cr^+ is just a mathematical object. The point of the construction is that Cr^+ is meant to offer a perspicuous representation of our scientist's belief state, which we initially represented by Cr . The idea is that the scientist's model of the world really does have an extra parameter. She sees the world not just as a Humean pattern of outcomes, but as a pattern that is "governed" or "generated" by

11 In its non-revolutionary ambition, my proposal is in the same spirit as [Skyrms 1980], [Skyrms 1984], [Jeffrey 1983], and [Blackburn 1983].

12 Skyrms [1980; 1984] calls this a "pragmatic" reduction of chance. Unlike a semantic reduction, a pragmatic reduction offers no informative analysis of chance statements. 'There is a 50% chance of heads' is not reducible to a statement about outcomes. But we can reductively explain what it means to *believe* that there's a 50% chance of heads, or to give such-and-such credence to that hypothesis. Beliefs about chance are reducible to other aspects of a rational belief state.

stable chances.

Why would a scientist have such a model of the world? It might be a cultural artefact, a holdover from a time when people thought that the physical world is governed by the commands of some deity. But I suspect there is a deeper reason. As I'll explain in section 7, having an extra chance parameter is computationally useful. This situation is well-known in the study of Artificial Intelligence: when reasoning about a given feature space, it is often useful to extend the space with extra parameters and perform reasoning in the extended space. I suggest that scientific reasoning takes place in the extended space of Cr^+ , although it is in principle reducible to reasoning in the original space of Cr .

All this was stage 1. So far, I've only talked about credence. We have other attitudes towards chances: one may *hope*, or *fear*, or *desire* that some outcome has a high chance. How are these attitudes to be explained?¹³ I also haven't looked at the language of chance: how should we understand the meaning of chance statements, and the role of such statements in communication?

I'll turn to language in a moment. First, a quick note on the other attitudes. Suppose an agent's beliefs are defined over a space of "worlds" with extra parameters or dimensions. By default, we should expect that the agent's other propositional attitudes may also involve these parameters: an agent who can believe that p can also hope or fear or desire that p . This is reflected in the standard Bayesian model of [Jeffrey 1983], where credence and desirability are defined on the same algebra.

Following Jeffrey, let's assume that the desirability function V is governed by the *Averaging* principle

$$V(A \vee B) = V(A) Cr(A | A \vee B) + V(B) Cr(B | A \vee B), \quad \text{if } Cr(A \wedge B) = 0.$$

Let's also assume that our scientist has no intrinsic desire for particular chance values: given full information about outcomes, she doesn't care about the chance with which these outcomes came about. The extended desirability $V^+(w, \theta)$ of an enriched world (w, θ) then doesn't depend on θ . By the Averaging axiom (applied to the extended desirability function V^+ and the extended credence function Cr^+), it follows that the extended desirability $V^+(A)$ of a proposition A (with $Cr^+(A) > 0$) is the Cr^+ -expectation of $V(w)$ conditional on A , where w ranges over possible sequences of outcomes. Since Cr^+ is reducible to Cr , it follows that $V^+(A)$ is determined by V and Cr . It is not determined by V alone.¹⁴

5 The Language of Chance

Let's turn to language. How should we understand the meaning of chance statements, if there are no chances out in reality?

One might think that we can't give a truth-conditional semantics. In discussions of realism and antirealism, it is sometimes claimed that truth-conditional semantics explains the meaning of sentences in terms of what has to be the case, in objective reality, for the sentence to be true. Some even suggest that truth-conditional semantics specifies the objective conditions that would *make* the relevant sentence true (see, e.g., [Schroeder 2023: p.30]). An antirealist, arguing that sentences about moral-

¹³ In metaethics, this is known as the "many attitudes problem"; see [Schroeder 2023: pp.89f.].

¹⁴ It is possible to write out $V^+(A)$ as a function of V and Cr , but this requires some technical machinery; I don't think it would lead to much illumination.

ity or chance aren't in the business of describing objective reality, must then reject truth-conditional semantics. The standard alternative is to go "expressivist": meaning is explained not in terms of truth-conditions, but in terms of attitudes canonically expressed by uttering a sentence.

In this vein, we might suggest that 'there is a 50% chance of heads' canonically expresses a certain credal state towards outcomes: a state that regards the relevant sequence as exchangeable and whose de Finetti representation contains a parameter μ that is concentrated on $1/2$. We'd then have to explain what, say, negated chance statements might mean. Presumably these, too, would express a credal attitude. If we want a uniform account of negation, we'll have to become expressivists about negation in general – and, for parallel reasons, about conjunction, conditionals, modals, attitude reports, and any other construction that can embed statements about chance.

There are honourable attempts to develop such a globally expressivist semantics. But they have found virtually no uptake in the field of natural-language semantics, where the truth-conditional approach has delivered a wealth of insights and results in comparison to which the fruits of expressivist accounts look meagre indeed.

As an alternative to expressivism, some have suggested going "fictionalist" (see, e.g., [Kroon et al. 2019]). In one version, fictionalism holds that statements that appear to carry problematic commitments contain hidden elements that cancel the commitments. In this vein, we might suggest that 'Radium-227 has a half-life of 42 minutes' means that *according to the fiction of chances*, radium-227 has such a half-life. In another version, fictionalism holds that the problematic statements are typically not asserted, but only "quasi-asserted", in more or less the way in which one quasi-asserts 'A dragon once lived in a teacup' when telling a story – without really believing what one says. Both proposals face serious problems. When we make statements about chance, we certainly aren't aware that we are engaging in some kind of pretence. Nor is there any linguistic evidence that chance statements are tacitly prefixed by a commitment-cancelling operator.¹⁵

Fortunately, there is no need to go expressivist or fictionalist, once we get clear about the metaphysical implications of truth-conditional semantics. As [Lewis 1975] emphasized, the ultimate point of associating linguistic expressions with meanings (referents, truth-conditions, or whatever) is to systematize facts about how language is used: how people use words to convey beliefs, ask questions, make requests, and so on.

The details can be spelled out in different ways. We don't need to follow Lewis's particular proposal. By way of illustration, let me briefly review another influential model, going back to [Stalnaker 1978].¹⁶ On Stalnaker's model, every conversation unfolds against a set of propositions, the *common ground*, that are taken for granted by the participants: all conversational participants accept the common ground, all accept that they all accept it, and so on. To a first approximation, *accepting* a proposition means treating it as certain for the purpose of the conversation.¹⁷ An utterance of a declarative sentence has the function of adding a corresponding proposition to the common ground.

¹⁵ See, e.g., [Rosen and Burgess 2007] for worries along these lines, and [Kroon et al. 2019: ch.8] for other objections to fictionalism.

¹⁶ Lewis and Stalnaker both follow [Grice 1957] in assuming that meaning facts are grounded in facts about intentional states. There are also views on which they are grounded in subpersonal mental states (e.g. [Chomsky 1980, 2000]); such accounts are even more favourable to antirealism.

¹⁷ See, e.g., [Stalnaker 2002], [Mandelkern and Dorst 2022], [Dinges 2024] for discussion and refinements.

Which proposition is added depends on which sentence is uttered. The task of semantics is to specify this association between sentences and propositions.

In this context, a *proposition* is a candidate object of acceptance or belief. For a variety of reasons, these are commonly modelled as sets of possible worlds. But what sort of “worlds” are we talking about?

Suppose the participants in a conversation have a world model with extra parameters: the doxastic possibilities they distinguish are in some respect more fine-grained than metaphysical possibilities. (In other respects, they may be more coarse-grained.) If we model their attitudes as sets of worlds, the relevant worlds will be enriched worlds with extra parameters. Since the common ground is defined in terms of belief, the propositions in the common ground will likewise be sets of enriched worlds, with extra parameters. What we need from semantics, then, is a mapping from sentences to sets of “worlds” with the extra parameters. These are the *truth-conditions* associated with a sentence.

So we don’t need to re-invent semantics. Chance statements can get a straightforward truth-conditional interpretation: ‘The chance of heads is $\frac{1}{2}$ ’ is true at a world-with-extra-chance-parameters iff the world’s chance parameter for heads has value $\frac{1}{2}$. On the linguistic side, everything can remain just as it is for the realist.

In some respect, assertions about chance do play a very different role than assertions about outcomes. This becomes clear when we examine the un-extended credence function Cr of speakers and hearers. If your extended credence is sure that a coin is biased towards heads, your un-extended credence is skewed towards worlds with more heads than tails, and this skew is insensitive to evidence about individual outcomes. An utterance of (1) conveys this resilient skew.

- (1) The chance of heads is greater than $\frac{1}{2}$.

If I accept your utterance and update my extended credence function accordingly, my un-extended credence function Cr will display the same type of resilient skew. It will do so without having been conditionalized on any proposition about outcomes: no previously possible outcomes have been ruled out.

Here we see an advantage of having a language with expressions for chance. Chance statements can be used to convey sophisticated patterns in the underlying credence function. It is unclear how this could be done without expressions for chance. One could, of course, say things like (2).

- (2) I regard the sequence of outcomes as exchangeable and my credence in any sequence strictly increases with the ratio of heads to tails.

But this is not just cumbersome. It is also too specific. Beliefs about chance can rely on less obvious symmetries in the underlying credence function than exchangeability. Chance statements conveniently abstract away from these details. Worse, (2) is a report about the speaker’s beliefs. If you utter (2), and I accept what you say, I will come to believe that you have a credence function of the form you described. I may or may not adopt such a credence function myself. Unlike (1), (2) does not have the canonical effect of transferring the relevant feature of your credence function to mine.

6 Objectivity

A perennial worry about antirealist views is that they can't capture the felt objectivity of our judgements. If the chances are mere projections of our attitudes, don't they depend on us? Surely the half-life of a lone radium-227 atom in a distant galaxy is what it is, no matter what you or I might think.

The account I have outlined respects this intuition. Judgements about chance are not analysed as judgements about our attitudes. Sentences like (3) and (4) that express a dependence of chances on attitudes are predicted to be false.

- (3) If we change our beliefs about chance, the chances will change.
- (4) If our beliefs about chance were different, the chances would be different.

The account is a form of quasi-realism, in the sense of [Blackburn 1984: ch.5]: it accepts almost all the judgements that one might have thought only realists could accept. We can agree that the half-life of radium-227 does not depend on us, that we can be wrong about it, and that we may never know it. Only in philosophical discussion does the difference between the two views come to light. The quasi-realist effectively has two models of the world. They have an enriched model with extra parameters, but they also suspect that these parameters are not a primitive part of the world *an sich*.

There's another way to press the objectivity worry. Are disagreements about chance genuine disagreements? If you and I happen to have different symmetries in our underlying credence functions, we'll end up with different beliefs about chance. Given that the symmetries can't be true or false, how can we say that at least one of us is objectively wrong? Of course, we can *say* it: if I believe that the half-life of radium-227 is 42 minutes, and you that it is 42 years, I will say that you are wrong. But if we deny the existence of objective chances, shouldn't we recognize, at least in a philosophical context, that our disagreement is not a genuine, substantive disagreement?

Response: it is not so easy for rational agents to have radically different symmetries in their credence functions. If we haven't looked at any relevant data, we had better not have any firm views about the half-life of radium-227. If, on the other hand, we have encountered the same evidence about (say) decay frequencies, and we have both applied Bayes' Rule, we can be expected to arrive at similar symmetries in our credence functions, even if we started with different priors. You'd have to start with very wacky priors to end up with an extended posterior credence in which the half-life of radium-227 is 42 years.

The probability calculus alone doesn't rule out wacky priors. My response therefore assumes that the norms of rationality go beyond probabilistic coherence. And they do.¹⁸ A rational prior should favour simple worlds with regular patterns, and it should give approximately equal credence to equally simple and regular worlds. If such priors are conditionalized on sufficiently rich evidence, they will generally converge to similar views about chance.

I can't prove this claim. The proposed rationality constraints are notoriously difficult to pin down.¹⁹ And I obviously haven't *shown* that there are substantive (non-formal) constraints on rational priors.

¹⁸ Some have even argued that there is a unique rational prior; see [Kopeck and Titelbaum 2016].

¹⁹ There are well-known "convergence theorems" showing that Bayesian agents who assign prior credence 0 to the same

(De Finetti himself disagreed.) All I suggest is that *if* we accept substantive constraints on rational priors, irresolvable disagreements about chance won't easily arise.

7 Bounded Rationality

Let's take a closer look at the symmetries required to derive de Finetti's Theorem. In section 3, I assumed that our scientist's credence over outcomes is invariant under permutation. This can be weakened. De Finetti's Theorem has been generalized to "partial exchangeability" and further to invariance under an abstract class of outcome transformations.²⁰ But we need *some* substantive symmetry, some pattern of conditional independence, in the underlying credence function. What justifies this assumption?

One might argue that the symmetries are a requirement of rationality: they are implied by the requirement that a rational prior is indifferent between "similar" cases, given that similarity is invariance under the right transformations.

I am sympathetic to this idea. But we don't need its full strength. It's enough if credence functions actually display the required symmetries, and that this is rationally permitted. (It need not be obligatory.) And there is a good reason why we should expect to find the required symmetries in human credence functions.

Consider a sequence of 50 coin flips. There are $2^{50} = 1,125,899,906,842,624$ possible results. How could an agent's cognitive system store a credence over these possibilities? A naive approach would require a database with 1,125,899,906,842,624 entries, each specifying a particular sequence and its probability. When the agent acquires information about the actual sequence – say, that the first outcome is heads – all the numbers in the database would have to be revised.

This is not a feasible approach. It would require an excessive amount of memory, and updates would be dismally slow. How else could one store (and update) a probability measure over the possible sequences?

If the probability measure that we want to store regards the sequence as exchangeable, the task becomes much simpler. We only need to store a distribution μ over a variable θ , from which one can compute the probability of any particular sequence as $\int_0^1 \theta^{\#h} (1 - \theta)^{\#t} \mu(d\theta)$. If the distribution μ is chosen from the versatile family of beta distributions, this requires storing just *two* numbers: the parameters α and β of the beta distribution. (We need three numbers if we also want to store that there are 50 coin flips. But even 3 is a big improvement over 1,125,899,906,842,624.) Updating also becomes much easier. If we learn that the first outcome is heads, we simply add 1 to α .

Of course, there are other ways to store and update probabilities over binary sequences. But all feasible options involve substantive symmetries. The credal symmetries required for de Finetti's Theorem are forced upon us by our cognitive limitations. A cognitively limited agent who wants to give positive credence to a wide range of possible patterns of outcomes must have a credence function that is compressible into a manageable form; that compression brings about the required symmetries.

events and who have the same likelihood model (in de Finetti's terms, whose priors display the same symmetries) will "almost surely" reach the same posterior when faced with sufficiently much data. This might help, but of course we can't assume fixed likelihood models. See [Earman 1992: ch.6] for further discussion.

20 See [Diaconis and Skyrms 2017: ch.7] or [Poirier 2011] for a summary.

It doesn't matter if the symmetries are required or even permitted by the norms of ideal rationality. They are required by the norms of bounded rationality.

If this is correct, it puts the whole gambit into a different light. De Finetti showed that a credence Cr with certain symmetries is representable as a credence Cr^+ that is unsure about a chance parameter θ . The present line of thought suggests that our beliefs aren't just *representable* in de Finetti's form. They must be *represented* in some such form in our cognitive architecture. The de Finetti representation (or some variant of it) is "cognitively real". It matches the format in which bounded agents store their beliefs.

We don't need de Finetti's Theorem to argue this point. We can point out directly that an efficient and versatile way to store and update a probability function over outcomes represents it as a mixture of chance hypotheses.

Sidelining de Finetti's Theorem brings certain advantages. For example, it eases the pressure to derive all our statistical practice from symmetry assumptions. I've assumed that this can be done. But it is, in fact, not a settled matter.

Here is one complication. Suppose we observe a sequence of binary outcomes. The first 100 outcomes are regular alternation of H and T: HTHT...HT. Faced with this evidence, I would suspect that this is not an entirely random sequence. I would be more confident that the next outcome is H than that it is T. If the first 100 outcomes had been THTH...TH, I would have been more confident that the next outcome is T. It follows that my prior credence didn't regard the sequence as exchangeable: HTHT...HTH had greater prior probability than THTH...THH, even though the two sequences are permutations of one another.

Intuitively, if we don't have any information about a sequence, we should divide our credence between different hypothesis classes: one class models each outcome as a (non-trivial) independent chance event; another models the sequence as following a deterministic pattern (like HTHTHTHT...); another models it as a Markov chain in which the next outcome probabilistically (and non-trivially) depends on the previous outcome.

The resulting credence function will have few symmetries. As far as I can tell, none of the generalizations of de Finetti's Theorem would apply. We might try to find the required symmetries in the credence function conditional on a particular hypothesis class, but it's not obvious how the conditional symmetries we now require could be cashed out purely in terms of credences over outcomes.²¹

This is a technical problem. No doubt the credence functions I've described will have some characteristic feature in common. The real question is why a scientist's (actual or idealized) credence function should have this feature. Why should a rational credence function be decomposable into a mixture of credence functions that are sure about chances?

The current stage of my proposal answers this question. A limited agent must store their beliefs about outcomes in a compressible form. There's no good reason why the cognitive representation must treat the outcomes as independent chance events. That's *one* easy representation. Another is to assume that the outcomes form a simple deterministic pattern. Another is that they form a Markov chain. A mixture of these representations is still computationally tractable, and has a better

²¹ One problem: if we want to identify hypothesis classes with sets of sequences, where does the alternating sequence HTHT...HT go? If it doesn't go to the "independent chance event" class, the credences conditional on that class won't be exchangeable.

chance of capturing the patterns in the world. Non-ideal rationality demands having such a mixed representation. We don't need to worry about how it could be derived from independently motivated symmetry considerations.

I'm almost done. One last stage. I don't think our brain actually stores and updates beliefs in the way I've described. We are not that smart. More likely, our brain uses some messy approximation to the format I've described – or rather, to the generalization of this format that is needed once we acknowledge that the world is not a sequence of coin flips. The approximation will still involve chance parameters. Those parameters *should* be determined by beliefs about outcomes, as implied by Bayes' Rule and the Principal Principle, and by the ultimate purpose of the parameters: to simplify reasoning about complex patterns of outcomes. But there is no guarantee that they are determined in this exact way. An agent's beliefs about chances may fail to fully align with their beliefs about outcomes.

In this final stage, Bayes' Rule and the Principal Principle emerge as rational norms. A confused agent may violate them. A confused agent may not even be probabilistically coherent. I suspect that real people are all somewhat confused in this way.

In the final stage, the chance parameters in our world model have become free: they can vary (at least to some extent) independently of beliefs about outcomes. This may explain the appeal of realism. If our representation of the world involves free parameters for chance, it is no surprise that we should be tempted to believe that the world itself contains irreducible chance facts. The reality of chance might be an illusion, created by the way our brain stores and updates information about a chance-free reality.

8 Outlook

De Finetti's gambit is often presented as a mathematical theorem combined with a revolutionary philosophical comment: "probability does not exist!" [de Finetti 1970]. I've tried to develop this approach into a more comprehensive philosophical theory. I've argued that, for largely computational reasons, our representation of the world involves extra parameters for chance that don't track an independent feature of reality. Scientific and proto-scientific reasoning operates on this representation. So does communication about chance.

Semantically, the view I have described agrees with primitive propensity theories: chance statements can't be informatively analysed in more basic terms. Unlike primitivist theories, however, the view can explain our statistical practice. It explains why the primitive chance parameters should be probabilistically linked, in a very precise way, to observed sequences of outcomes.

Metaphysically, my proposal is Humean, insofar as it doesn't assume any modal elements in objective reality. Unlike standard Humean accounts, it does not attempt to translate chance statements into statements about the Humean mosaic. Such translations not only tend to be intrinsically implausible (see [Schwarz 2018]); they also threaten to contradict our statistical practice.

Throughout, I have relied on the simplifying assumption that the world is a sequence of coin flips. Dropping this assumption would lead to a much richer picture. The full picture would clarify the deep connections between chance, projectibility, and resiliency, discussed in [Skyrms 1980: ch.IA], [Skyrms 1984: ch.3], and [Skyrms 1994]. It would clarify the role of causal propensities in Bayesian

networks (see, e.g., [Pearl 2000: ch.7]), and connect with the “arbitrary functions” approach to non-fundamental chance (see, e.g., [von Plato 1983], [Strevens 2003], [Myrvold 2016]). I have to leave these developments for another occasion.

To conclude, let me take a step back and consider how the proposed account might generalize beyond the case of chance. Could we develop, say, an antirealist account of morality on the present model?

I can’t go into any details, but the prospects look promising. Antirealist accounts in metaethics are usually construed as hypotheses about moral language. Moral assertions are given a special kind of meaning or communicative function. This requires sweeping revisions of established linguistic theories. Even if the revisions are successful, further difficulties remain. The antirealist has to make sense of our attitudes towards morality: how can we *believe* that eating people is wrong, or *desire* that it is wrong, or be *unsure* whether it is wrong, if ‘eating people is wrong’ doesn’t express a proposition?

We can bypass these problems by starting with credence. In the case of chance, this was a natural starting point because the best-understood aspect of chance – perhaps the only well-understood aspect – is how chance figures in Bayesian reasoning. A metaethical position that follows our path would start with moral credence. It would explain how and why our credences are defined over an enriched space of propositions with extra moral parameters. Models of this kind are familiar from [Gibbard 1990: ch.5] and the subsequent literature. What hasn’t yet been spelled out fully is how an agent’s credence function over the enriched space is determined, how it responds to evidence, how it interacts with the agent’s desires (and possibly other attitudes), and how it affects the agent’s behaviour. This is where the main action would lie. The functioning of moral language might largely take care of itself.

Would this be a better approach than standard, language-centric approaches? I don’t know. But it seems worth exploring.

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